

```
[5]: def RATROW():  
      name='vijay'  
      for i in range(len(name)):  
          for j in range(i+1):  
              print(name[i],end='')  
          print()  
RATROW()
```

```
v  
ii  
jjj  
aaaa  
yyyyy
```

```
[7]: def RATCOL():  
      name='vijay'  
      for i in range(len(name)):  
          for j in range(i+1):  
              print(name[j],end='')  
          print()  
RATCOL()
```

```
v  
vi  
vij  
vija  
vijay
```

```
[8]: def ILATROW():
      name='vijay'
      for i in range(len(name)):
          for j in range(len(name)-1,i,-1):
              print(' ',end='')
          for k in range(0,i+1):
              print(name[k],end='')
          print()
```

ILATROW()

```
    v
   ii
  jjj
 aaaa
yyyyy
```

```
[10]: def ILATCOL():
       name='vijay'
       for i in range(len(name)):
           for j in range(5,i,-1):
               print(' ',end=' ')
           for k in range(0,i+1):
               print(name[k],end='')
           print()
```

ILATCOL()

```
    v
   vi
  vij
 vija
vijay
```

```
[12]: def PRTCOL():  
    name='vijay'  
    for i in range(len(name)):  
        for j in range(5,i,-1):  
            print(' ',end=' ')  
        for k in range(0,i+1):  
            print(name[k],end=' ')  
        print()  
PRTCOL()
```

```
    v  
   v i  
  v i j  
 v i j a  
v i j a y
```

```
[13]: def PRTRROW():  
    name='vijay'  
    for i in range(len(name)):  
        for j in range(len(name)-1,i,-1):  
            print(' ',end=' ')  
        for k in range(0,i+1):  
            print(name[i],end=' ')  
        print()  
PRTRROW()
```

```
    v  
   i i  
  j j j  
 a a a a  
y y y y y
```

```
[15]: def RATUPPERROW():  
        name='vijay'.upper()  
        for i in range(len(name)):  
            for j in range(0,i+1):  
                print(name[i],end=' ')  
            print()
```

RATUPPERROW()

```
V  
I I  
J J J  
A A A A  
Y Y Y Y Y
```

```
[18]: def RATUPPERCOL():  
        name='vijay'.upper()  
        for i in range(len(name)):  
            for j in range(0,i+1):  
                print(name[j],end=' ')  
            print()
```

RATUPPERCOL()

```
V  
V I  
V I J  
V I J A  
V I J A Y
```

```
[19]: def IRATUPPERROW():  
        name='vijay'.upper()  
        for i in range(4,-1,-1):  
            for j in range(5,i,-1):  
                print(' ',end='')  
            for k in range (0,i+1):  
                print(name[i],end=' ')  
            print()
```

IRATUPPERROW()

```
Y Y Y Y Y  
A A A A  
J J J  
I I  
V
```

```
[21]: def IRATUPPERROW():  
        name='vijay'.upper()  
        for i in range(4,-1,-1):  
            for j in range(4,i,-1):  
                print(' ',end='')  
            for k in range (0,i+1):  
                print(name[k],end=' ')  
            print()
```

IRATUPPERROW()

```
V I J A Y  
V I J A  
V I J  
V I  
V
```

```
[23]: def IPRTCOL():  
    name='vijay'  
    for i in range(4,-1,-1):  
        for j in range(5,i,-1):  
            print('',end=' ')  
        for k in range(0,i+1):  
            print(name[k],end=' ')  
        print()
```

IPRTCOL()

```
v i j a y  
 v i j a  
  v i j  
   v i  
    v
```

```
[24]: def IPRTROW():  
    name='vijay'  
    for i in range(4,-1,-1):  
        for j in range(5,i,-1):  
            print('',end=' ')  
        for k in range(0,i+1):  
            print(name[i],end=' ')  
        print()
```

IPRTROW()

```
y y y y y  
 a a a a  
  j j j  
   i i  
    v
```

```
[25]: def fact():  
      num=int(input('Enter the number:'))  
      for i in range(1,num+1):  
          if num%i==0:  
              print(i)
```

fact()

Enter the number: 90

1
2
3
5
6
9
10
15
18
30
45
90

```
[26]: def biggest(a=10,b=60,c=30,d=50):  
      name=lambda a,b,c,d:a if a>b and a>c and a>d else b if b>c and b>d else c if c>d else d  
      print(name(a,b,c,d))
```

biggest()

60

```
[30]: def smallest(a=10,b=20,c=100,d=200):  
      name=lambda a,b,c,d:a if a<b and a<c and a<d else b if b<c and b<d else c if c<d else d  
      print(name(a,b,c,d))  
  
      smallest()
```

10

```
[31]: num=[8,4,7,21,43,98,70,76,54,53,13]  
      group=list(map(lambda a:a**3,num))  
      print(group)
```

[512, 64, 343, 9261, 79507, 941192, 343000, 438976, 157464, 148877, 2197]

```
[32]: def group():  
      num=[2,4,6,7,8,3]  
      root=list(map(lambda a:a**2,num))  
      print(root)  
  
      group()
```

[4, 16, 36, 49, 64, 9]